

CHAPTER 8

Cortical interactions in vision and awareness: hierarchies in reverse

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Abstract: The anatomical connections between visual areas can be organized in ‘feedforward’, ‘feedback’ or ‘horizontal’ laminar patterns. We report here four experiments that test the function of some of the feedback projections in visual cortex. Projections from V5 to V1 have been suggested to be important in visual awareness, and in the first experiment we show this to be the case in the blindsight patient GY. This demonstration is replicated, in principle, in the second experiment and we also show the timing of the V5-V1 interaction to correspond to findings from single unit physiology. In the third experiment we show that V1 is important for stimulus detection in visual search arrays and that the timing of V1 interference with TMS is late (up to 240 ms after the onset of the visual array). Finally we report an experiment showing that the parietal cortex is not involved in visual motion priming, whereas V5 is, suggesting that the parietal cortex does not modulate V5 in this task. We interpret the data in terms of Bullier's recent physiological recordings and Ahissar and Hochstein's reverse hierarchy theory of vision.

Introduction

The classical view of visual cortical processing is one of simple receptive fields in area V1 distributing information to higher areas in which cells have larger receptive fields and more complex response properties. A cell in V1, for example, may have a receptive field (RF) of less than one degree of visual angle and give its largest response to a bar oriented within a few degrees of a critical orientation (Hubel and Wiesel, 1977). By comparison, a cell in extrastriate area V4 may have a RF size up to 30 degrees or more and respond to more complex configurations of shape and color (Heywood and Cowey, 1987). At the next stage in visual cortex, the inferotemporal cortex, one finds neurons that respond selectively to faces, objects and

different views of the preferred stimulus (Gross et al., 1972). This bottom up hierarchical view is still dominant. It is true that we now know more about back projections from higher to lower areas and about parallel processing of visual attributes. However, the interactions between higher and lower areas are usually considered in two contexts. First, in a bottom up constructivist manner in which the lower areas feed the higher areas with simple information that is then elaborated; second in a top down, modulatory manner in which higher areas influence the selectivity of lower areas. The weight of evidence for this view depends partly on the kinds of questions that have been asked of V1 neurons. When illusory contours were considered, and indeed termed

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‘cognitive contours’ it was presumed that the perception was generated in a cognitive, higher area such as inferotemporal cortex. Subsequent studies of V2(von der Heydt et al., 1984) and later V1 neurons (Grosf et al., 1993), however, showed that these lower areas contained the necessary architecture and response properties to retrieve the form of illusory contours. Similarly, because attention, in one of its many flexible guises, was considered to be dominated by inferotemporal and parietal cortices, few experiments were aimed at assessing whether a V1 or an *extrastriate* neuron changed its responses to stimuli depending on behavioral relevance. Those that did probe these and other questions aimed at reassessing the role of these areas in visual processing consistently found that V1 neurons were indeed sensitive to the context of a visual scene (Zipser et al., 1996; Nothdurft et al., 1999), showed responses that can be described as attentional (Motter, 1993; Somers et al., 1999; Brefczynski and DeYoe, 1999), and respond at times later than the mean latencies for secondary visual areas (Schmolesky et al., 1998; Bullier, 2001). Further, a new influential theory of vision provides principled reasons for reconsidering the role of V1 in vision (Ahissar and Hochstein, 2000). In their reverse hierarchy theory of vision, Ahissar and Hochstein propose that visual processing follows a global to local trajectory in which extrastriate neurons with large RFs carry out an initial coarse-grained analysis of the visual field followed by a more detailed analysis in earlier visual areas.

Against this background we have used transcranial magnetic stimulation (TMS) to examine interactions between visual areas. In this paper we consider the interactions between areas V5 and V1 in awareness (Covey and Walsh, 2000; Pascual-Leone and Walsh, 2001), between extrastriate and striate cortex in visual search (Juan and Walsh, 2003) and between the parietal cortex and extrastriate cortex in visual priming (Campana et al., 2002). Based on the results of these experiments we argue that V1 is necessary for visual awareness, that feedback to V1 is necessary for accurate performance in detecting visual targets in a noisy array, and that extrastriate area V5, rather than the parietal cortex, is the critical site for priming of visual motion. These results are consistent with the reverse hierarchical view of vision.

In the discussion we place these results in a wider context.

Experiment 1. V1 and awareness: neuropsychological evidence

Rationale

One of the areas of contention in the role of V1 in higher functions concerns its putative role in awareness. One view is that activity in an extrastriate area specialized for processing a given visual attribute is sufficient for awareness of that attribute (Zeki, 1993). An alternative view (Stoerig and Covey, 1997; Weiskrantz, 1997) is that activity in an extrastriate region only contributes to awareness as a result of interaction with V1. Both these positions have been supported in experiments on the blindsight patient GY, but it is always problematic to compare visual experiments on GY because subtle differences in stimulus conditions can result in large differences in the results. By using TMS one can bypass the geniculostriate visual pathways in GY and induce activity directly into V5. If this activity is sufficient for awareness, GY should report seeing moving visual phosphenes when TMS is applied over his V5.

Methods

We compared the effects of TMS over V1 and V5 in GY, in a 61 year old, peripherally blind patient, PS (Covey and Walsh, 2000), and in six normal subjects. TMS was applied using a Magstim Super Rapid stimulator with a 70 mm figure of eight coil. TMS was delivered either in single pulse mode or at 4, 8, 10 or 12 Hz for a maximum of 500 ms over V1 or V5. During TMS subjects sat fixating the center of a polar plot drawing board at a distance of 57 cm. PS ‘fixated’ by placing an index finger at the center of the board. Following each pulse or train of pulses the subject reported whether or not a phosphene was seen. If so they were asked to draw/describe the sensation. Care was taken to ensure that the site was not missed by TMS and following co-registration of the coil with GY’s structural and functional MRI scans twenty five points were stimulated in several attempts to elicit phosphenes.

Results

As Fig. 1 shows, when TMS was applied over V1 of the control subjects, PS, or the intact hemisphere of GY phosphenes were elicited and showed the intact cortices to be organized retinotopically. When TMS was applied over V5 of the six control subjects or PS, moving phosphenes were reported. In GY, however, moving phosphenes could only be elicited by stimulating over the right, intact hemisphere. When TMS was applied over left V5 it was impossible to elicit the perception of moving phosphenes in the contralateral visual field.

Figure 1

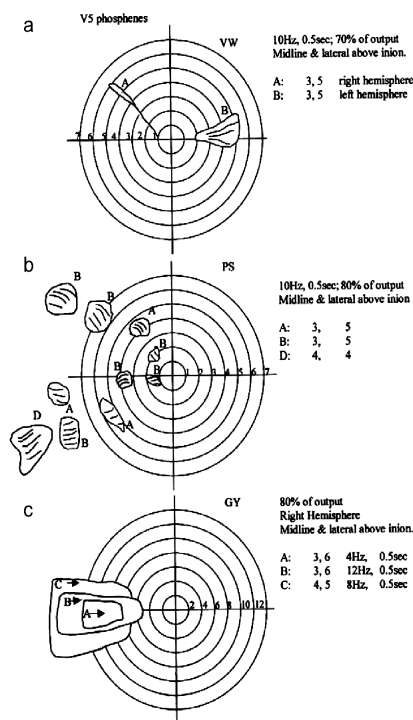


Fig. 1. Moving phosphenes generated by applying TMS to V5 of a neurologically intact subject (a), a subject with bilateral optic tract damage and therefore no input to the striate cortex (b) and a subject with damage to the striate cortex of the left hemisphere (c). In GY it was not possible to elicit phosphenes by stimulating left V5.

Conclusion

From this experiment we inferred that activity in extrastriate area V5 alone was insufficient to produce awareness of motion. However, this experiment did not address directly the question of interactions between V1 and other areas and it is this we turn to next.

Experiment 2. V1 and awareness: neurochronmetric evidence

Rationale

If, from the results of the above experiment, one can conclude that backprojections from V5 to V1 are critical for visual awareness, it follows that this projection must have a time course.

Methods

TMS was applied over two visual cortical sites, V5 and V1, using a combination of phosphene generation and disruption methods in eight subjects. We first established that subjects perceived moving phosphenes following TMS over V5 and stationary phosphenes, in a region of the visual field that overlapped with the perceived moving phosphenes with TMS over V1. Subjects then received suprathreshold TMS over V5 such that they reliably perceived movement and this was either preceded or followed by TMS over V1 delivered at 80% of phosphene threshold. Thus the V1 TMS was insufficient to mask a V5 generated percept with a V1 generated percept. The V5-V1 TMS asynchrony was between -100 and -10 ms in steps of 5 ms (i.e. V1 TMS preceded V5 TMS) or between +10 and +100 ms in steps of 5 ms (i.e. V1 TMS followed V5 TMS). Subjects were asked to state whether they perceived a phosphene and if so whether it was moving, possibly moving but they were uncertain, present but stationary or whether there was no phosphene at all.

Results

When TMS over V1 occurred prior to TMS over V5 there were no changes in subjects' reports of phosphene perception. When V1 TMS followed V5 stimulation there were clear changes in perception (fig. 2). Disruption of V1 processing between 5 and 45 ms after activity induced in area V5 resulted in fewer reports of motion and in some subjects a complete absence of the perception of movement.

Conclusion

These data show that the backprojection from V5 to V1 is necessary for visual awareness of activity in V5 (as shown by Experiment 1) and that the time course

of the interaction covers a window of between 5 and 45 ms after signals reach V5. In the two experiments described so far, we have 'bypassed' V1 either by using subjects without input to V1 or with damage to it (Experiment 1) and also by directly stimulating V5 and in effect exciting V5 neurons by what might be considered an abnormal route. This of course leaves open the question of how V1 might operate in more natural circumstances and in Experiment 3 we address this.

Experiment 3. V1 and complex scene discrimination

Rationale

The implications of the reverse hierarchy theory of vision (Ahissar and Hochstein, 2000) extend to many

Figure 2

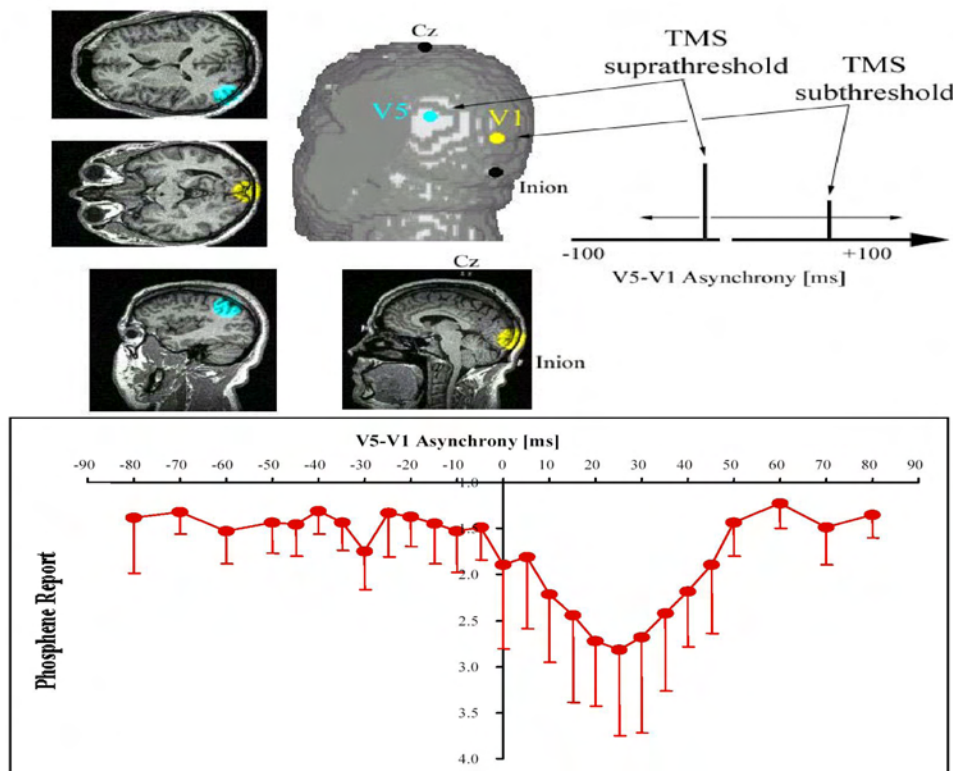


Fig. 2. TMS over V1 before V5 has no effect on the percepts induced by V5 TMS. But V5 TMS that is followed by V1 TMS disrupts the perception of the induced phosphene perception of motion.

functions other than awareness and one of the most challenging hypotheses is that V1 is important as a look-up table for the fine details of the visual scene (Lee et al., 1998; Lamme and Spekreijse, 2000). If this were the case it should be possible to apply TMS to V1 at later times in processing and to disrupt visual discrimination performance.

Methods

Feature and conjunction visual detection tasks were used in the experiments. Subjects reported the presence or absence of a target defined either by a unique color (either a blue circle amongst red circles or a red circle amongst blues) or a unique combination of color and orientation (a blue forward slash amongst an array of blue back slashes and red forward slashes). On a trial in which the target was present there were 11 distracters and on a target absent trial there were 12 nontargets. The search array subtended 2x2 degrees of visual angle. Subjects viewed the stimuli binocularly, 57 cm from the screen. Each trial began with a 500 ms central fixation point followed by the search array and a visual mask. The mask was displayed until subjects made the present/absent response by a key press. Subjects were asked to respond accurately without any time pressure. Performance was manipulated by varying the stimulus-mask onset asynchrony (SOA) in steps of 10 ms in a staircase procedure until performance reached between 62.5 and 87.5% correct. The mean SOA in the feature task was 50 ms (range 30-100) and in the conjunction task 120 ms (range 80-180 ms). Responses were subjected to signal detection analysis (Green and Swets, 1966). In each experiment there was a block of 160 feature trials and a block of 160 conjunction trials. TMS was delivered randomly on half the trials.

Three experiments were carried out to test this hypothesis. In Experiment 3a subjects received rTMS at 10 Hz for 500 ms at the onset of the visual stimulus in order to establish whether V1 disruption would have any effect on either of the visual search tasks. In Experiment 3b subjects received rTMS at 10 Hz for 500 ms with the onset of rTMS beginning 100 ms after the onset of the visual array. This was to test whether the first 100 ms, if left unstimulated, would be sufficient for V1 to carry out all the processing required in feature and conjunction tasks.

In Experiment 3c subjects received double pulse TMS (dTMS) to try to establish if the role of V1 in feature search was confined to a particular range of time within the first 100 ms after stimulus onset and to test whether the role of V1 in conjunction search had any temporal specificity. In the feature search task, which Experiment 2 showed to involve V1 only in the first 100 ms, subjects received dTMS at 0 and 40 ms, 0 and 100 ms and 80 and 120 ms after the onset of the visual array. In the conjunction search task which, as Experiment 2 showed, requires V1 involvement outside the first 100 ms (we could not say, without Experiment 3, whether the first 100 ms is also important) subjects received dTMS at the same times as in the feature task and also at 140 and 180, 200 and 240 and 260 and 300 ms after stimulus onset.

A Magstim Super-Rapid TMS machine with a 70 mm figure of eight coil was used. Stimulation intensity was set at 65% of stimulator output. The coil was centered approximately 1 cm above theinion, selected on the basis of phosphene production or scotomas induced in previous experiments. We have shown previously that stimulation over this region principally excites V1 (Cowey and Walsh, 2000; Pascual-Leone and Walsh, 2001: see also Kammer, 1999).

Results

Experiment 3a

Repetitive pulse TMS significantly impaired performance on both the features (d' decrease 0.8: $t(7)=2.27$, $P<0.05$, Fig. 3a) and the conjunction tasks (d' decreased by 0.8: $t(7)=3.01$, $P<0.05$, Fig. 3b).

Experiment 3b

There was no effect of TMS on the feature search task when the first 100 ms after the onset of the visual array were left unstimulated ($t(5)=0.02$, $P>0.05$, Fig. 3c). In the conjunction task rTMS reduced d' by 0.4 ($t(5)=3.07$, $P<0.05$, Fig. 3d).

Experiment 3c

In the feature task dTMS produced a significant cost in d' ($F(3, 15)=6.26$, $P<0.01$, Fig. 3e) when applied

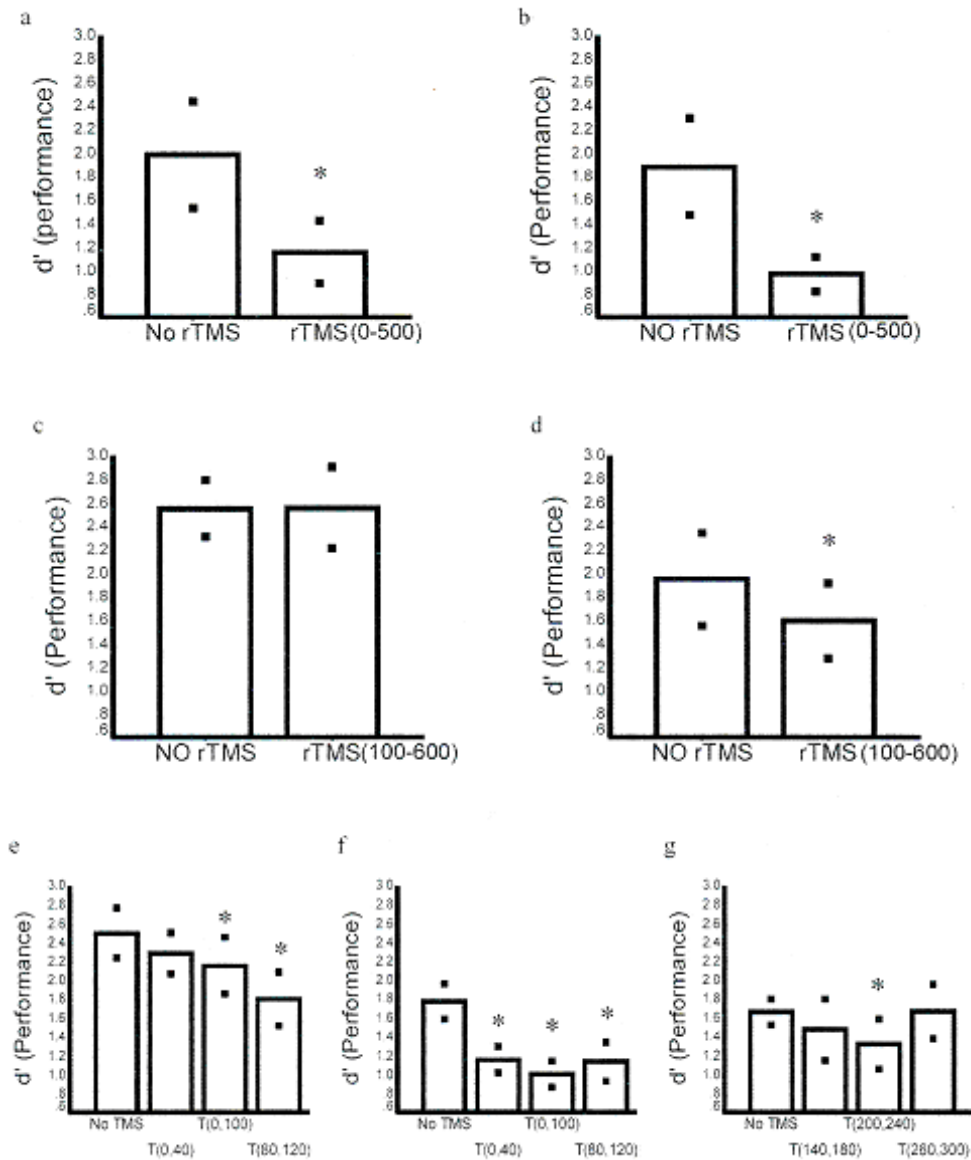


Fig. 3. The effects of TMS over V1 on feature and conjunction d' scores. (a) Feature search, (b) conjunction search (c) Feature search, (d) conjunction search (e) Feature search, (f) and (g) conjunction search.

at 0 and 100 ms ($t(5)=2.85$, $P<0.05$) and at 80 and 120 ms ($t(5)=6.05$, $P<0.01$) after array onset. In the conjunction task dTMS significantly impaired performance ($F(3, 15) = 10.98$, $P<0.01$, Fig. 3f) when applied at any of the three early intervals 0, 40($t(5)=3.84$, $P<0.05$); 0, 100 ($t(5) = 5.09$, $P<0.01$); and 80,120 ms ($t(5)=14.42$, $P<0.01$)) and one of the

later intervals (200, 240 ms ($t(5)=1.72$, $P=0.053$, Fig. 3g).

Conclusion

The interference with V1 for 500 ms after visual stimulus onset shows V1 to be required in both visual

feature and conjunction tasks (Experiment 3a). Delaying the onset of rTMS for 100 ms after visual stimulus onset shows that V1 is only required in the first 100 ms for feature search, but is important beyond the first 100 ms in conjunction search (Experiment 3b). Applying dTMS shows V1 to be involved in feature search in the later part of the first 100 ms and in the conjunction task throughout the first 100 ms and again between 200 and 240 ms (Experiment 3c). These findings thus establish that V1 is necessary for normal performance in conjunction visual search at times that suggest a sustained or recurring involvement of V1 as visual analysis proceeds. We interpret this as further evidence supporting the reverse hierarchy view of vision. Hierarchies are a feature of all cortical processing and in addition to exploring the view that V1 may have a critical position in the visual sensory hierarchy we have examined the role of V5 in what might be called a 'visuocognitive' hierarchy. It is sometimes argued that extrastriate areas are modulated by the parietal cortex. However, the evidence for this is indirect and in experiment 4 we tested this assumption directly.

Experiment 4. The visual cortex and priming

Rationale

In the previous three experiments we explored interaction between V5 and V1 but there are many assumptions about the interactions between areas at different levels of the visual hierarchy that can be challenged. One such assumption concerns the visual abilities of the right posterior parietal cortex which has been associated with visual priming. It has long been contested, however, that extrastriate areas are critical to short-term memory (Martin-Elkins et al., 1989) and also to priming of visual attributes (Bar and Biederman, 1999; Walsh et al., 2000). The theoretical basis for this like the reverse hierarchy theory, presents a challenge to the orthodoxy that so-called higher functions require so-called higher areas. Tulving and Schacter (1990) have articulated the Perceptual Representation System (PRS) view of memory for visual attributes and Magnussen and Greenlee (1999) have proposed a similar view of what they term perceptual memory based on

psychophysical studies. In this experiment we tested the hypothesis that a secondary visual area with some relative specialization for a visual attribute would be the area likely to represent short-term changes in the likelihood of that attribute being an important component of the visual scene, and therefore important for priming/short-term memory. The competing hypothesis is that the parietal cortex would be the area responsible for short-term priming of visual attributes (Marangolo et al., 1998).

Methods

Subjects were presented with four panels of 100% coherent motion. Each panel subtended 1.9° of visual angle and were separated from each other by 0.57° . Seven subjects, five male and two female, aged between 23 and 29 years and all right handed, participated in the motion priming condition. Five subjects, three male, two female, aged between 25 and 39 years and all right handed, participated in the color priming control. Five right-handed subjects three male and two female, aged between 25 and 31 years, participated in the third experiment to control for nonspecific response time (RT) decreases following TMS over a nonvisually related area, in this case the vertex. All subjects were of a sufficient level of scientific education to understand the information given about magnetic stimulation and gave written informed consent according to the Declaration of Helsinki. The experiment was approved by the local ethics committee, Oxford Research Ethics Committee.

Stimuli and procedure

Stimuli were generated using a PC Pentium II. 400 MHz processor and presented on a Sony CPD-G200 Trinitron flat screen monitor (800 x 600 pixels, refresh rate 120 Hz) driven by a Rage IIC AGP display adapter. Stimuli were four virtual squares ($1.9 \times 1.9^\circ$ of visual angle and black, like the background. 0.31 cd/m^2) regularly arranged around the center of the screen and separated by 0.57° from the two adjacent squares. Each square contained 100 spatially random white pixels (57.2 cd/m^2) all moving either horizontally or vertically at $2.96^\circ/\text{s}$ in the same

direction within a square. In three of the squares the direction of motion was identical, while in the fourth square it was orthogonal to the other three. This odd direction was the target to be detected by the subject. Each trial began with a white fixation point (0.2° , 57.2 cd/m^2) for 500 ms and followed by a blank screen. Stimulus conditions and procedures were identical for a color priming task, designed to test for nonspecific effects of TMS, with the exception that motion was in the same direction (top-bottom) in all four squares, the target was the odd color [either green target (CIE.288.597) and red distracters (CIE.609.352) or vice versa] and stimuli were presented for 48 ms because good performance on the color task required fewer frames than the motion experiment.

Subjects had to select one of four response buttons, using the thumb and first fingers of each hand, to indicate which square contained the target. The rTMS pulses were applied 500 ms after the response (i.e. in the second 500 ms of the intertrial interval) and were followed by the fixation spot for the next trial. A target was present on every trial. To avoid possible effects of spatial priming the target never appeared in the same square on any two consecutive trials, a fact of which the subjects were not informed and which they did not realize during the experiment. This manipulation also ensures that the priming data are not confounded either by spatial priming effects or repetition of finger responses. The direction of moving dots and location of the target was pseudorandomised between trials. In the motion experiment, subjects were given 320 trials in blocks of 40, and received 80 TMS trials per stimulation site (V1, V5/MT and PPC) and 80 control trials. In the color experiment subjects were given 240 trials in blocks of 40, and received 80 TMS trials per stimulation site (V5/MT and PPC) and 80 control trials. In any given block TMS was given either on every trial or none of the trials. In TMS experiments there is a choice between delivering TMS predictably or randomly. In our experience the best method depends on the experimental design and the task being performed. In this experiment we chose a block design to avoid an interaction between trials that were preceded by TMS or no TMS; this would have been theoretically uninformative in the context of our hypothesis, and would have required twice the number

of TMS and nonTMS trials. Practice blocks were allowed to ensure that the subjects could perform the task accurately and that the priming effect was present.

In the motion experiment, the independent variable was the direction of the doors in the target square. The direction could either be the same as or different from the previous trial, and the aim of the statistical analyses carried out was to assess the effects of TMS on the relative reaction times of trials in which the preceding target was moving in the same direction versus those trials in which the target was preceded by a different direction. In the color experiment, the color of the dots in the target square was the independent variable. Throughout we refer to these as Same and Different trials respectively.

A third experiment was carried out to ensure that any decreases in reaction time could be attributed to intercessory facilitation and not to the effects of pro-activating visual cortex by applying TMS over visually related areas, V1 V5/ MT and PPC during the intertrial interval. In this control experiment the visual stimuli and magnetic stimulation parameters were identical to those used in the motion experiment with the exception that the TMS was applied over the vertex. The subjects were stimulated by TMS and also by the sound of the discharge and the tactile sensation on the scalp. Conceivably subjects might respond more quickly because the TMS had a nonspecific alerting effect which could decrease reaction times to an irreducible level at which it is not possible to interpret differential effects of primed and nonprimed trials.

Magnetic stimulation

The stimulator was a MagStim 200 Super-Rapid stimulator delivering current to a 70 mm figure-of-eight coil. Details of the stimulator coil selection and the intensity of stimulation have been given elsewhere (Ashbridge et al., 1997). Repetitive pulse TMS was applied to three sites. For V5 /MT stimulation the coil was held tangential triangular to the skull with the handle pointing backwards at $\sim 90^\circ$ to the axis of the spinal cord and for parietal stimulation the

handle pointed backwards at $\sim 45^\circ$ to the spine. For V1 stimulation, the coil handle was oriented vertically. Pulses were delivered at an intensity of 60% of maximum output of the stimulator at 10 Hz for 500 ms.

The posterior occipital site was located between 2 and 4 cm above the inion at the midline. We shall refer to this site as striate cortex or V1. We are aware that it is unlikely that only V1 was stimulated, but the shorthand is justifiable on several grounds. First, the interpretation of the phosphene data is consistent with the topography of striate cortex (Kastner et al., 1998; Kammer, 1999). On behavioral grounds V1 is also the most frugal interpretation. Since single and repetitive pulse TMS produce effects consistent with interference to V1 whether one considers the time course of the effects (Pascual-Leone and Walsh, 2001) the phenomenology of stimulation on perceptual recognition (Kosslyn et al., 1999; Cowey and Walsh, 2000) or the disruptive effects of stimulation (Amassian et al., 1993). A left hemisphere lateral occipital site was located 3 cm dorsal and 5 cm lateral to the inion (individual Subjects' coordinates varied between 3 and 4 cm dorsally and 5 and 6 cm laterally). We refer to this site as V5/ MT on the basis of previous anatomical data showing that this site overlies extrastriate area V5/ MT and on the numerous previous studies that have selectively impaired visual motion perception with TMS over this site (Hotson et al., 1994; Pascual-Leone et al., 1999; Stewart et al., 1999). As in previous studies we designated the V1 and V5 /MT sites on the basis of phosphene production; the V1 sites was selected as the point of stimulation which gave the most consistent phosphene in the central few degrees of the visual field and the chosen V5/MT sites was where TMS produced a clear moving phosphene [for descriptions of how moving phosphenes are established and how they vary see (Stewart et al., 1999)]. To establish the appropriate parietal cortical site we used a visual search task as a behavioral assay, and the region at which Stimulation produced a deficit on this task was taken as the critical parietal location for visuospatial processing. The details of TMS and visual search have been given in several sources (Ashbridge et al., 1997; Walsh et al., 1998a, b; Walsh and Cowey, 1998).

Results

Motion priming

Baseline trials in the absence of TMS showed a clear priming effect — reaction times on Same trials were ~ 40 ms faster than on Different trials (paired sample t-test, $r=3.3$, $df=6$, $P=0.016$). A two-way analysis of variance (ANOVA) was carried out with factors stimulation site (three levels: V1, V5/MT and parietal) and priming (two levels: Same prime, Different prime). There was a significant effect of priming ($F=8.609$, $P=0.026$) and a significant interaction between priming and TMS stimulation site ($F=11.465$, $df=2$, $P=0.002$). TMS over V5 MT in the intertrial interval abolished the priming effect (Different trials were marginally, though not significantly, faster than Same trials), and post hoc contrasts showed that the interaction of TMS site and priming were due to the effects of TMS at V5/MT relative to V1 ($F=10.776$, $df=1$, $P=0.017$) and relative to the parietal site ($F=16.847$, $df=1$, $P=0.006$). Repetitive pulse TMS in the intertribal interval decreased reaction times across all conditions with no differences between V5/MT, V1, or parietal TMS induced RT decreases ($F=1.05$, $df=2$, $P=0.38$). Thus TMS over V5/MT had a specific effect on reaction time — it changed the relative differences between Same and Different trials. The overall decrease in reaction times has been seen in many TMS experiments and is an example of nonspecific inter-sensory facilitation. An elegant solution to the problem of evaluating this kind of speeding was presented by Marzi et al., (1998), who interpreted differences between facilitations as the real deficit caused by TMS. In our study this rationale is particularly relevant since the effect of interest was not merely a predicted difference between TMS stimulation sites but a predicted difference within (Same versus Different) and between (V1, V5/MT, parietal) sites. The effects of TMS were not due to an overall decrement in performance as measured by accuracy. A 3×2 ANOVA with factors stimulation site and prime showed that performance on Same trials was more accurate than on Different trials ($F=11$, $df=1$, $P=0.016$) but there was no significant effect of stimulation site ($F=0.201$, $df=2$, $P>0.05$) nor an interaction between TMS and accuracy on

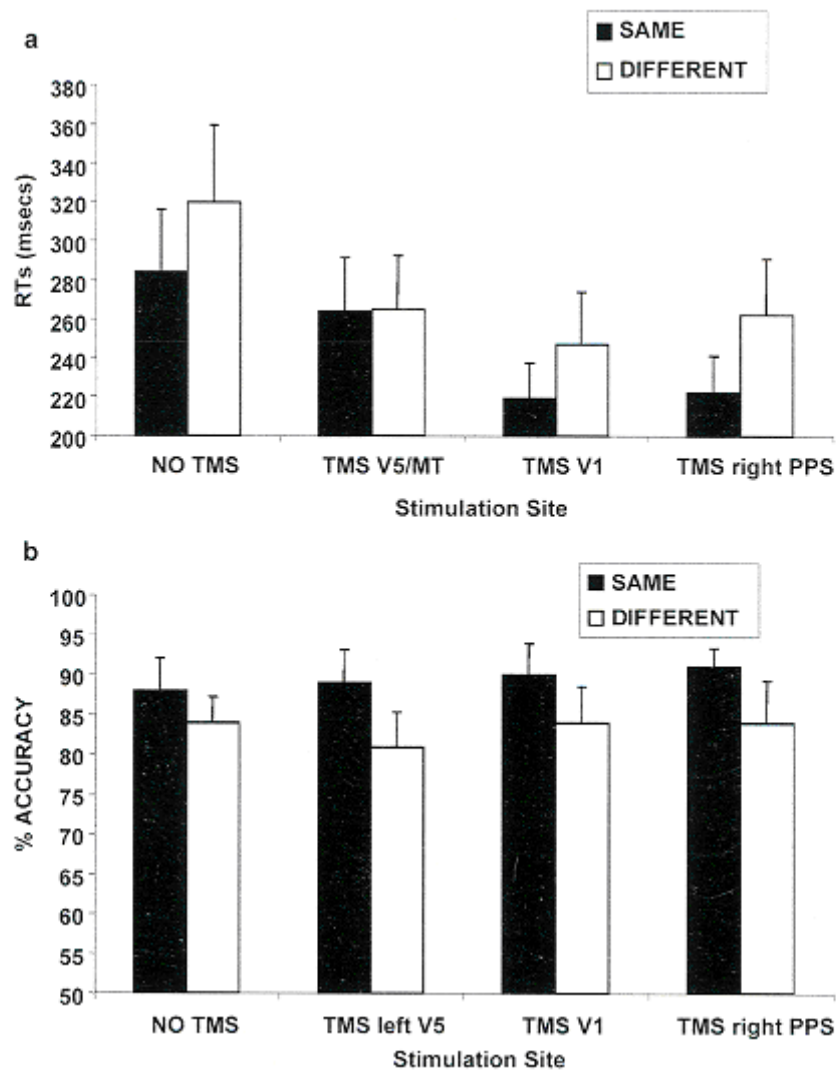


Fig. 4. The effects of TMS over V5, V1 and posterior parietal cortex on visual motion priming, (a) Only V5 TMS disrupts the RT advantage of being presented with a direction target presented on the previous trial. (b) Motion accuracy per se was unperturbed by TMS, even over V5.

primed and unprimed trials ($F=0.56$, $df=2$, $P>0.05$). Figure 4 shows the data for motion priming.

Color priming

The difference between the effects of TMS over V5 /MT, PPC and V1 provided controls for specificity of TMS stimulation site. A color condition was included to control for specific task demands to ensure that the effects of V5 /MT stimulation were specific to motion and not any priming task. Baseline

trials in the color priming condition also showed a clear priming effect in the predicted direction with a mean RT advantage of 30 ms for Same over Different trials ($t=5.55$, $df=4$, $P<0.005$). TMS over V5 /MT or parietal cortex had no effect on color priming. An ANOVA with prime (Same versus Different) and rTMS site (V5/ MT versus parietal) as factors showed the effect of the prime to be significant ($F=38.83$, $df=1$, $P<0.005$) but no effect of TMS site ($F=0.012$, $df=1$, $P>0.05$) and no interaction between TMS site and prime ($F=0.383$, $df=1$, $P>0.05$). A 3×2 ANOVA with factors stimulation site and

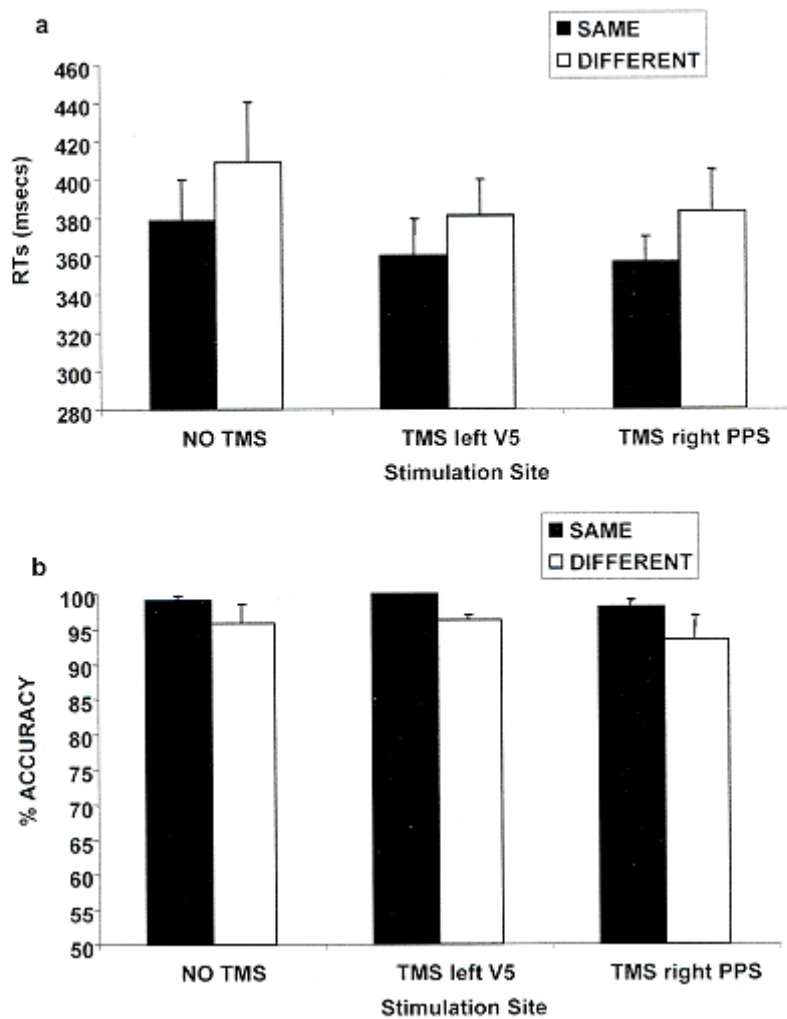


Fig. 5. TMS over V5 and PPC had no effects on color priming (a) or in color discrimination accuracy (b).

prime showed that performance on Same trials was slightly more accurate than on Different trials ($F=7.42$, $df=1$, $P=0.053$) and there was no significant effect of stimulation site ($F=1.3$, $df=1$, $P>0.05$) and no interaction between TMS and accuracy on primed and unprimed trials ($F=0.185$, $df=1$, $P>0.05$). In other words, the effect of TMS over V5 /MT was specific to motion. Figure 5 shows the data.

Vertex stimulation and motion priming

A two-way ANOVA (condition: TMS nonTMS; priming: Same/ Different) showed that the priming effect of Same direction was maintained ($F=26.34$,

$df=1$, $P=0.007$) when TMS was applied over the vertex to control for visually related reaction time enhancements. The reaction time enhancements caused by nonspecific intersensory effects of TMS were significant ($F=7.8$, $df=1$, $P=0.049$), and there was no interaction between the effects of prime and TMS ($F=3.06$, $df=1$, $P=0.155$), indicating that TMS over vertex did not interfere with motion priming.

Conclusion

The results of this experiment establish that visual area V5 is the region responsible for the short-term

priming of visual motion. Parietal cortex and primary visual cortex, on the other hand do not play a critical role in priming.

General discussion

There are two themes to this series of experiments. The first is to explore cortico- cortical interactions in vision. The second is to raise the question: Why do we always look to higher areas to explain higher functions. Why would we want to raise this question at all? Well, to quote the person honoured in this volume as soon as everyone believes something it is likely to be wrong. This dictum is more than mere mischief. If we are to decide a priori that all spatial and vaguely attentional functions require the posterior parietal cortex, that all memory functions require areas beyond the sensory cortex and that all back projections to the sensory areas are modulatory (in the top down sense), then we have made some brave assumptions about how the brain works. And if these assumptions are wrong, or at least greatly limited, as they seem to be, further progress will be hindered. Collectively the experiments reported here challenge these assumptions and are best interpreted within frameworks that are not yet standard. The finding that V1, or interaction between V1 and extrastriate cortex, is necessary for visual awareness, can be explained by Bullier and colleagues' physiological work and Ahissar and Hochstein's Reverse Hierarchy theory. The fourth experiment was conceived within and is explained by Tulving and Schacter's Perceptual Representation System hypothesis (akin to Magnussen and Greenlee's Perceptual Memory System).

We can look forward, then to extending the approach taken in these experiments but perhaps these too are limited --- particularly by what they have borrowed from standard approaches. While we have begun to explore the reverse hierarchy views, we have done so with old stimuli – bars, dots and colors presented briefly, in the belief that these impoverished conditions either adequately represent the visual environment or at least the building blocks of it. There are two important ways in which this faith in simple stimuli may be wrong, and both can be remedied by the use of natural images and the presentation of familiar stimuli. It is clear that the environment presents many regularities to the sensory

system and these can be described as statistical distributions of signals along dimensions such as wavelength, intensity, location, contrast, motion, time etc... and we can therefore investigate how the visual system encodes and responds to stimuli it may have evolved to handle (Olshausen and Field. 1996; Baddeley et al., 1997; Baddeley and Hancock. 1991). Using these kinds of stimuli and analytical methods may provide harsh tests of any theory based on simple arrays of bars colors and movement.

Using familiar stimuli is also an important next step to take. Consider most visual experiments; the subjects are naive to the strange stimuli, the simple response and the experimental setting. In our everyday vision, however, we tend to be very familiar with what we see and what we do with what we see. One tends to wake up in the same bedroom, take the same train, walk the same streets, lose the same diary on the same desk etc... we search, remember and act in somewhat constant surroundings. Novelty is, well novel and a pessimistic view of much of visual cognition might be that we have developed elaborate theories of how subjects respond to novel stimuli in a novel environment and in a novel way. It is not a given that these theories will translate to natural scenes and behaviors.

The reverse hierarchy theory presents a framework with which we can test the processing of natural images. The spatial scale of correlations in an image can provide a measure of what might be considered global and local in real scenes, from which we can extend the work reported in Experiment 3. Experiments 1 and 2 can also be extended using complex images. Scenes will contain correlations between attributes as well as within and this may provide a new starting point for the study of binding. Finally, what we remember from a scene may be changed greatly by competing information and it is important to evaluate the sensory memory hypothesis in the real world.

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